Concepts of Information and Entropy in Probability Theory and Shannon's Information Measure

March 15, 2026

Abstract

The lecture discusses foundational concepts in information theory, focusing on Shannon's groundbreaking idea of measuring information and entropy. The key learning objective is to understand how unexpected probabilistic events provide more information than routine ones. Specifically, the lecture demonstrates that the amount of information in an event is related to the event's probability: the less likely an event, the more information it carries. Mathematically, information is represented as the logarithm of the inverse of the probability. The lecture also introduces the concept of entropy as the average information content or uncertainty in a system, defined as the weighted sum of the information of all possible states. Entropy reaches its minimum value (zero) when the system's state is known with certainty and its maximum when all states are equally likely, thereby reflecting maximum uncertainty about the system.

1 Introduction to Information Content in Probabilistic Events

This section introduces a core concept in information theory — comparing the information content of two probabilistic events. Using a simple example, the lecture illustrates how the relative likelihood of events affects the information they provide.

Information Content and Probability

The information content of an event is inversely related to the probability of that event. Formally, if an event A occurs with probability $P(A)$, then the information content $I(A)$ is given by:

$$I(A) = \log\left(\frac{1}{P(A)}\right) = -\log\Big(P(A)\Big).$$

1.1 Illustrative Example

Consider two probabilistic events involving a professor's behavior:

- **Event 1:** Professor N enters the room.
- **Event 2:** Professor N expels a student from the room.

The goal is to determine which of these events provides more information.

Comparing Probabilistic Events

In general, rare events provide more informational content than common ones. Thus, the event with a lower probability will convey more information.

In this example, the first event (professor entering the room) is likely to be common, while the second event (professor expelling a student) is relatively rare. Therefore, the rarer event — the professor expelling a student — carries more information.

Likelihood vs. Information

If the probability of Professor N entering the room is P_1 and the probability of expelling a student is P_2 with $P_1 > P_2$, then:

$$I(\text{expelling a student}) > I(\text{entering the room}),$$

because $P_2 < P_1$.

This simple example sets up the fundamental relationship between the probability of an event and the amount of information it provides.

2 Shannon's Logical Framework for Information

In this section, we continue building on Shannon's logical steps that formalize the relationship between probability and information content. The key insight is that less probable events convey more information.

2.1 Uncommon Events in Practical Scenarios

To further illustrate the concept, let's examine situations where mass media might be interested in reporting certain events:

- **Event 1:** A professor lectures in the classroom.
- **Event 2:** A professor quizzes students unexpectedly outside of class.

While the first event is expected and routine, the second is uncommon and surprising. From the media's perspective, the second event is more "newsworthy." The reason is that less probable events attract more attention and convey more information.

Why Uncommon Events Matter

The second event is less likely (lower probability) and therefore provides more informational value. This directly connects to Shannon's foundational insight that low-probability events yield higher information content.

Thus, the idea is that an unexpected or uncommon scenario generates more information than a routine or expected one.

2.2 First Logical Step: Information and Probability

The first logical principle laid out by Shannon is simple: the amount of information an event carries is a function of the inverse probability of that event. If $P(E)$ is the probability of event E, then the information content $I(E)$ can be expressed as:

$$I(E) = f\left(\frac{1}{P(E)}\right).$$

Information as a Function of Probability

Formally, the information content of an event E is some function f of the inverse of its probability:

$$I(E) = f\left(\frac{1}{P(E)}\right).$$

This function typically takes the form of a logarithm, which ensures consistency with key properties of information (e.g., additivity across independent events).

2.3 Second Logical Step: Additivity of Independent Events

The second logical step involves understanding how information behaves when we consider multiple probabilistic events that are statistically independent.

- Suppose we have two statistically independent events E_1 and E_2.
- Their joint probability is given by $P(E_1 \cap E_2) = P(E_1) \times P(E_2)$.

Now, the combined information content of these two independent events should be the sum of the individual pieces of information.

Additivity of Information

For two independent events E_1 and E_2, the total information $I(E_1 \cap E_2)$ is additive:

$$I(E_1 \cap E_2) = I(E_1) + I(E_2).$$

This property of additivity is critical in choosing an appropriate function f. In fact, it drives the use of the logarithmic function.

Why Additivity Matters

The additivity property ensures that when two independent events occur together, the total information is just the sum of the information from each event. This aligns with intuitive expectations about independent events.

2.4 The Role of Logarithms in Information Measurement

The choice of a logarithmic function for f is not arbitrary. It satisfies the key properties discussed above:

1. **Monotonicity:** Less probable events should give more information.

2. **Additivity:** The total information for independent events should be the sum of the information from each event.

Therefore, we define the information content of an event E with probability $P(E)$ as:

$$I(E) = -\log P(E).$$

Definition of Information

The information content $I(E)$ of an event E with probability $P(E)$ is defined as:

$$I(E) = -\log P(E),$$

where the logarithm base is often 2 (for bits), e (natural logarithm), or 10, depending on the application.

This definition elegantly encapsulates the key requirements: less probable events yield higher $I(E)$, and independent events have additive information.

Implication of Negative Logarithm

Since $-\log P(E)$ increases as $P(E)$ decreases, this definition perfectly aligns with our intuition: the rarer the event, the more surprising it is, and thus the more information it provides.

2.5 Additivity of Information for Independent Events

A key logical conclusion is that the information content for two independent probabilistic events is the sum of their individual information contents. In other words, if events E_1 and E_2 are independent, then the joint information is:

$$I(E_1 \cap E_2) = I(E_1) + I(E_2).$$

Additivity of Information

Theorem 2.1. *If two events E_1 and E_2 are independent, the total information content of their joint occurrence is the sum of their individual information contents:*

$$I(E_1 \cap E_2) = I(E_1) + I(E_2).$$

This crucial property leads us to the conclusion that the only mathematical function which satisfies the additivity condition for probabilities is the logarithmic function. This is not only intuitive but can also be formally proven.

Uniqueness of the Logarithmic Form

Theorem 2.2. *The only function f that satisfies both the monotonicity and additivity conditions for information content is the logarithm. Specifically, if:*

$$f(P(E_1 \cap E_2)) = f(P(E_1)) + f(P(E_2)) \tag{1}$$

for independent events E_1 and E_2, then $f(x)$ must be proportional to $\log(x)$.

Thus, the definition of information content as the logarithm of the inverse probability is not only intuitive but also mathematically unique under these constraints.

2.6 Choice of Logarithmic Base

Traditionally, natural logarithms (base e) have been used for convenience in mathematical derivations. However, if we instead use a base-2 logarithm, the units of information are measured in *bits*—corresponding to the familiar binary representation in computing.

Logarithm Bases and Information Units

- Natural logarithms yield information in units called *nats*.
- Base-2 logarithms measure information in *bits*.

While the choice of base depends on context, the core properties of information remain the same.

2.7 Definition of Entropy

Now we expand this notion of information to an entire probability distribution over a system with multiple states. Suppose a system can exist in n distinct states, and the probability of finding the system in state i is P_i.

We now define the *entropy* H of the system as the *expected value* of the information content across all possible states.

Entropy of a System

Definition 2.1. *Let a system have n possible states, each with probability P_i. The entropy H of the system is defined as:*

$$H = -\sum_{i=1}^{n} P_i \log P_i.$$

Entropy represents the average uncertainty or "information content" of the system. It reaches its maximum when all states are equally likely, and its minimum (zero) when the system is entirely predictable (i.e., when one state has probability 1 and all others zero).

Intuition Behind Entropy

Entropy quantifies the uncertainty of a probability distribution:

- If all states are equally likely, uncertainty is maximized.
- If one state dominates the probability (near certainty), uncertainty is minimal.

Shannon introduced an extremely interesting measure, which we now recognize as entropy. Essentially, entropy is the average information about the system — or more precisely, the average uncertainty remaining about the system's actual state.

Let us break down this concept into two key ideas.

1. **Information Content of Individual Events:** We already discussed this part. When an unlikely event occurs, we gain more information (since it's surprising), while a likely event tells us less.
2. **Overall Average Information (Entropy):** Instead of looking at single events, we can consider the average information over all possible states. That's what entropy is

— the weighted sum of the information for each possible state, where the weights are the probabilities of those states occurring.

Recap on Individual Event Information

We previously saw that the information content of a single event i with probability P_i is $-\log P_i$. Rare events produce more information than common events.

Now, Shannon wanted a measure that summarizes the overall "spread" of a probability distribution over all possible states. That measure is given by the entropy formula we introduced earlier:

$$H = -\sum_{i=1}^{n} P_i \log P_i.$$

It turns out that this concept of entropy is foundational not just to communication theory, but also to various fields including thermodynamics and machine learning. It helps us understand the average uncertainty of the system.

2.8 Mutual Information

The mutual information between two random variables measures how much knowing one of them reduces uncertainty about the other. It is a critical tool in quantifying the shared information between variables and plays a fundamental role in both coding theory and machine learning.

Definition of Mutual Information

The mutual information $I(X;Y)$ of two random variables X and Y is defined as:

$$I(X;Y) = H(X) + H(Y) - H(X,Y),$$

where $H(X)$ and $H(Y)$ are the individual entropies of X and Y, and $H(X,Y)$ is the joint entropy.

Mutual information can also be expressed as:

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

Intuition

If two variables are independent, knowing one gives no information about the other, so $I(X;Y) = 0$. If they are perfectly dependent, knowledge of one completely determines the other, maximizing $I(X;Y)$.

2.8.1 Relationship to Conditional Entropy

Conditional entropy is a measure of the remaining uncertainty of a random variable after knowing another variable. Mutual information helps connect joint entropy and conditional entropy in insightful ways.

Conditional Entropy

The conditional entropy of variable Y given X, denoted $H(Y|X)$, is defined as:

$$H(Y|X) = H(X,Y) - H(X).$$

It quantifies the amount of uncertainty remaining about Y when X is known.

Mutual information also has an equivalent definition in terms of conditional entropy:

$$I(X;Y) = H(Y) - H(Y|X) = H(X) - H(X|Y).$$

Interpretation

This means that mutual information tells us how much of the entropy (or uncertainty) of one variable is reduced by knowing the other.

Example: Coin Toss

Suppose X and Y represent the outcomes of two fair coin flips. If the coins are flipped independently, knowing X tells us nothing about Y. Thus,

$$I(X;Y) = 0.$$

However, if we knew that every time X landed on heads Y also landed on heads (and tails correspondingly), then $I(X;Y)$ would be maximized because knowing one immediately determines the other.

2.9 Kullback–Leibler (KL) Divergence

The Kullback–Leibler divergence (or KL divergence) is a measure of how one probability distribution differs from another reference probability distribution. Unlike mutual information, KL divergence is not symmetric, and it's often used as a measure of inefficiency when using an approximation for a true distribution.

KL Divergence

The KL divergence between two probability distributions P and Q over the same random variable is defined as:

$$D_{\mathrm{KL}}(P||Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right).$$

Interpretation

- If $P = Q$, the KL divergence is zero.
- KL divergence is always non-negative.
- Intuitively, it measures the "extra" bits of information required when the distribution Q is used to approximate P.

2.10 Information Gained from Observing a System State

When we observe a system and identify it in a particular state, we gain a certain amount of information about that system. Let's say the system can exist in different states, each with a probability p_i.

- If you observe the system and find it in a specific state (say the first state), you gain information regarding the system.
- The amount of information obtained depends on how likely or unlikely that specific state is.
- Less probable states convey more information when observed, as they are unexpected. Conversely, highly probable states convey less information.

The measure of this quantity of information gained after observing a specific state is the negative logarithm of the probability of that state, i.e.,

$$I = -\log p_i.$$

Understanding Information

The information content of an event tells us how "surprising" or "unexpected" that event is. The less likely an event, the more information it provides once it occurs.

2.11 Uncertainty Before Observation

On the other hand, before you measure or observe the system, you only know the probabilistic distribution of possible states but do not yet know the specific state the system is in. In such a case, the system remains "mysterious" to you. You have partial knowledge, i.e., you know the probabilities, but not the actual outcome.

This situation of partial knowledge leads to uncertainty. The question now becomes: how do we measure this uncertainty?

- If you knew the specific state beforehand, your uncertainty would be zero.
- However, because you only know the probabilities and have no observation yet, you have some degree of uncertainty.

Uncertainty as a Measure

We need a measure of how incomplete your knowledge is, i.e., how uncertain you are about the system's specific state prior to observation.

2.12 Entropy as a Measure of Uncertainty

This measure of uncertainty that arises from a set of known probabilities (but unknown actual outcomes) is known as **entropy**. The concept of entropy quantifies the average uncertainty in a probability distribution over the possible states of a system.

In the next section, we will mathematically define and discuss how entropy measures the degree of uncertainty present when only probabilistic information is known, and no specific measurement has been taken yet.

- **Minimal Entropy:** Entropy achieves its minimal value, which is zero, when one of the states has a probability of 1, and all other states have a probability of 0.
- In other words:

$$p_1 = 1 \quad \text{and} \quad p_i = 0 \quad \text{for} \quad i \neq 1.$$

This scenario signifies complete certainty about the system's state. With absolute certainty, no uncertainty remains, and hence, entropy reduces to zero. This outcome is straightforward because knowing exactly which event will occur implies no new information is gained at the moment it happens.

Interpretation of Zero Entropy

When entropy is zero, the system's future state is completely predictable. This indicates that the "measure of uncertainty" is entirely absent. You already know the outcome with complete confidence, so no new information is gained when the event occurs.

2.13 Maximum Entropy and Uniform Distributions

Conversely, entropy reaches its maximum value when all states are equally likely. Suppose a system can exist in n possible states, and each state has equal probability $\frac{1}{n}$. The entropy for such a uniform distribution is:

$$H = \log n.$$

High Entropy

When all states are equally likely, there's maximum uncertainty about which state will occur. This general condition leads to maximal entropy, as every event carries an equal "surprise factor" or information content.

2.14 Conclusion: Role of Entropy in Information Theory

Entropy serves as a foundational concept in information theory. Its value reflects how much uncertainty (or lack of knowledge) exists about the system's state before observation.

Final Insight

Entropy enables us to quantify how much "information" we expect to gain from learning the outcome of a probabilistic system. A lower entropy indicates that the outcome is more predictable (less information is gained), while higher entropy suggests greater unpredictability and thus higher information potential.

Summary of Key Points

- Entropy measures the average uncertainty of a system's state.
- Minimal entropy ($H = 0$) occurs when one state has probability 1 (complete predictability).
- Maximal entropy occurs when all states are equally likely, i.e., a uniform distribution.

- Entropy quantifies the expected information gain from observing a system.

Further Reading

Shannon's original work on information theory provides a rigorous foundation for understanding entropy, uncertainty, and information generation. Relevant sections in standard texts on information theory can deepen your comprehension of these concepts.